Does a careful-comparison instruction improve multiple-choice accuracy?

A paired, fixed-model study on 200 ARC-Easy test questions, using constrained JSON output and local inference.

Model	Data	Evaluation unit
Qwen3-0.6B / Ollama 0.40.5	ARC-Easy test split	One question; two paired responses

Main finding

A / basic



B / careful comparison



Effect (B − A)	95% paired bootstrap CI	Exact two-sided McNemar
−2.0 percentage points	−5.5 to +1.5 pp	p = 0.38769531

Interpretation: A scored 161/200 (80.5%) and B scored 157/200 (78.5%). The comparison did not establish an accuracy benefit from the careful-comparison instruction. The result does **not** demonstrate equivalence or rule out effects smaller than this study can resolve.

Study completion	Inferential conclusion
PASS – 400 recorded answers; archive re-scored offline	**INCONCLUSIVE** – no demonstrated directional improvement

Evidence status

The submitted archive was checked offline against source hashes, deterministic sample selection, exact request payloads, all 400 raw answers, stored grades, and reported statistics. This is artifact-level verification, **not** independent third-party validation or independently witnessed model execution.

No additional main-study model calls were made during the archive review.

Study design and locked protocol

Research question

Does an instruction asking the model to carefully compare all options change its multiple-choice accuracy relative to a basic selection instruction, under otherwise fixed conditions?

Intervention

Variant	Frozen instruction
A — basic	Choose the single best answer to the multiple-choice question.
B — careful comparison	Read the question carefully and compare every option before choosing the single best answer.

Common instructions and controlled factors

The item question and original option labels/order were identical in both variants; options were displayed one per line. Every prompt ended with the same instruction to return a single JSON **answer** field, with no explanation. A dynamic JSON schema restricted that value to the item's existing labels.

Control	Frozen setting
Inference	Local Ollama **0.40.5**; model **qwen3:0.6b**; digest saved with every response
Generation	temperature 0; seed 42; think false; stream false; num_predict 32; num_ctx 2048
Ordering	Alternating A→B and B→A by sample index (100 pairs each order)
Scoring	Full response must parse as JSON with only **answer**; exact match to original answerKey

Data, sampling and pilot separation

The ARC-Easy **test** snapshot contains 2,376 questions. A deterministic SHA-256 ranking of question IDs with preselected seed **R01-ARC-EASY-TEST-20261019-42** selected 200 without replacement. A 15-question **validation** pilot informed the output-format design before the main sample was frozen; no pilot items overlapped the test items by ID or normalized question/choices.

The local protocol files were hashed before main-run execution. There was no external registration/timestamp or outside custodian; pre-run chronology is supported by the supplied local artifacts only.

Paired outcomes and uncertainty

Observed paired responses

Paired status	Questions	Share of 200
Both variants correct	153	76.5%
A correct; B incorrect	8	4.0%
A incorrect; B correct	4	2.0%
Both variants incorrect	35	17.5%

Outcome and uncertainty

Statistic	Result	Interpretation
A accuracy	161 / 200 = 80.5%	Wilson 95% CI: 74.46% to 85.39%
B accuracy	157 / 200 = 78.5%	Wilson 95% CI: 72.30% to 83.63%
Paired B − A	−2.0 pp	12 discordant pairs (8 A-only; 4 B-only)
Paired bootstrap	95% CI: −5.5 to +1.5 pp	10,000 resamples of 200 question-pairs; seed 42
Exact McNemar	two-sided p = 0.38769531	Not significant at preselected α = 0.05

What can — and cannot — be inferred

The observed 2-point difference favors A on this sample, but it is small relative to paired uncertainty. The interval includes both zero and a modest positive B effect. Therefore, a claim that B increases accuracy is unsupported; so are claims that A is generally better or that the two prompts are equivalent.

The statistical analysis counts **questions** as sampling units; the two responses to the same question are kept together. Separate Wilson intervals are descriptive and must not be substituted for the paired comparison.

Operational outcome

All 400 stored responses were in the required JSON schema, with 400 logged physical attempts and zero recorded unresolved transport errors. From the first to last recorded answer the run spanned about **18 minutes**; local hardware performance is not a general model benchmark.

The JSON constraint changes the construct being tested: this study measures **answer selection under enforced structured output**, not spontaneous compliance with a free-form formatting instruction.

Archive verification and reproduction

Archive-level checks completed

Verification step	Result
Hash links: snapshot, pilot, protocol, selection, lock, runner and run manifest	PASS
Recompute deterministic 200-item sample from the 2,376-row snapshot	PASS
Reconstruct every request from frozen instruction, question and schema	PASS — 400/400
Parse raw model outputs and re-score from gold labels	PASS — 400/400
Recalculate the paired 2×2 table, McNemar test and bootstrap CI	PASS

Reproduce the numerical analysis without Ollama

The reviewer package includes the original evidence ZIP and a portable, standard-library Python audit script. Run in any clean directory with Python 3.9+ (no model download and no extra packages):

Command
python verify_report01_archive.py Report01_Evidence_v1.zip

A PASS verifies that the original archive still contains the expected hashed files, frozen requests and matching recorded answers, and that the published summary is correctly calculated. It **does not** replay the model or independently establish who/when produced the model responses.

Key SHA-256 identifiers

Item	SHA-256 (abbreviated; full values in archive)
Frozen protocol	46b8692a96cf7a8f…c8a310267d2cf9
Frozen 200-item sample	b1d7412e00135fd6…759f0b07f0ef
Source test snapshot	0116e5edbed73991…21ae16a790
Main run manifest	76cae79818448be5…288a51b78d12b5
Original evidence ZIP	360e867cdb6af2c0…60f81ecbb8ca

Reproduction caveat

The archived inference and original audit scripts use a fixed Windows home-directory project path. The portable verifier removes that path dependency for **offline numerical reproduction** only. Fresh inference still requires installing the matching model/runtime and carrying out a separately controlled replay.

Limits, decision rights and next steps

Six-field assurance card

Field	Recorded statement
Risk	A small prompt change may shift answer accuracy; format success might be mistaken for answer correctness.
Use / scope	Exploratory model/prompt evaluation for ARC-Easy-style multiple-choice tasks; no autonomous deployment decision.
Evidence	200 paired test items, frozen instructions, 400 raw responses, archive hashes, paired inference.
Stop / qualification	No transfer to other models or tasks; no superiority/equivalence claim at p = 0.388; preserve null result.
Decision	No demonstrated improvement from the careful-comparison instruction under this exact setup.
Owner / review	Portfolio study author; request external clean-directory reproduction and critique before claiming outside validation.

Limitations that must remain visible

One small model, one public benchmark and one local run per prompt; possible training-data contamination; constrained output; limited power for small effects; locally recorded hashes rather than independent custody; no external reviewer. Main-run repeatability across machines, Ollama versions or model implementations was not tested.

Publication and licensing

The AI2 ARC dataset card identifies **CC BY-SA 4.0**. Preserve attribution to the source and assess share-alike/redistribution obligations before publishing datasets or examples; review locally embedded paths and metadata in the evidence ZIP before public release.

Recommended next step

Release the English report with a concise methodology README and the offline verifier; seek one person outside the project to reproduce the summary from a clean directory, then record their actual scope of review. Continue to Report #2 without altering Report #1 findings.

References

[1] AI2 ARC dataset and card (license, split, citation): huggingface.co/datasets/allenai/ai2_arc.
[2] Clark et al., “Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge” (2018), arXiv:1803.05457.
[3] Ollama structured outputs: docs.ollama.com/capabilities/structured-outputs.
[4] Primary evidence: user-supplied **Report01_Evidence_v1.zip**; portable reproduction script supplied alongside this report.

This study is an independent project by its author; it has not been independently validated by a third party.